Welfare with Product Returns: Sufficiency of Aggregates for Local Incidence

Draft notes

July 19, 2026

Abstract

With returns, gross demand $D(p)$ alone is no longer the right local welfare statistic. For first-order (local) incidence, aggregates $D(p)$ and $r(p)$ suffice: $dCS = -D(p)(1 - r(p))\,dp$. For second-order (global) deadweight loss, the full joint distribution of consumer types is required.

1 The problem

Without returns, aggregate demand is a sufficient statistic for local consumer surplus:

$$dCS = -D(p)\,dp.$$

With returns, a change in list price p moves both the extensive margin (who buys) and the intensive margin (each buyer's return probability). The aggregate return rate $r(p)$ is only the average among current buyers; it does not decompose how much of Δr comes from new entrants versus inframarginal behavioural change. So one cannot mechanically treat (D, r) as a black-box welfare object without a microfoundation.

2 The conjecture

Conjecture: $D(p)$ and $r(p)$ are *not* sufficient for welfare under returns; the joint distribution of individual return behaviour is needed, except in knife-edge cases (e.g. return probabilities independent of p, or shifting at a common rate).

3 The result

The conjecture is half right. It depends on the order of the welfare effect.

First-order (local) incidence. $D(p)$ and $r(p)$ *are* sufficient. Drop the pre-purchase value v and work with hassle cost $\kappa_i \geq 0$ and post-purchase match $u \sim G_i$. After buying at p, the consumer keeps if $u \geq p - \kappa_i$. Expected surplus from buying is

$$s_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

Buy iff $s_i(p) \geq 0$; keep probability $k_i(p) = \Pr(u \geq p - \kappa_i)$; return probability $r_i(p) = 1 - k_i(p)$.

Aggregates:

$$D(p) = \int \mathbf{1}\{s_i(p) \geq 0\}\, dF(i),$$
$$K(p) = \int \mathbf{1}\{s_i(p) \geq 0\}\, k_i(p)\, dF(i) = D(p)\big(1 - r(p)\big).$$

Total consumer surplus $CS(p) = \int \mathbf{1}\{s_i \geq 0\}\, s_i\, dF$. Differentiate and apply the envelope theorem (marginal buyer has $s_i = 0$, so the extensive margin drops out):

$$CS'(p) = -\int \mathbf{1}\{s_i \geq 0\}\, k_i(p)\, dF(i) = -K(p) = -D(p)\big(1 - r(p)\big). \tag{1}$$

Hence for a small price change,

$$dCS = -D(p)\big(1 - r(p)\big)\, dp.$$

Composition effects cancel at first order because the marginal buyer adds zero surplus. Aggregates (D, r) appear directly on the right-hand side.

Second-order (global / DWL). The conjecture is fully correct. Curvature $CS''(p)$ (e.g. a Harberger triangle) depends on how the composition of buyers changes with p, which requires the joint distribution of (κ_i, G_i). Aggregates (D, r) alone cannot identify it. That is beyond the scope of local pass-through / incidence analysis.

4 Addendum for the incidence draft

Consumer side (sufficient microfoundation). Each consumer i has hassle cost κ_i and post-purchase match $u \sim G_i$. At list price p,

$$s_i(p) = \mathbb{E}_{u \sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

Buy if $s_i(p) \geq 0$; keep probability $k_i(p) = \Pr(u \geq p - \kappa_i)$. Gross demand and kept units:

$$D(p) = \int \mathbf{1}\{s_i(p) \geq 0\}\, dF(i),$$
$$K(p) = \int \mathbf{1}\{s_i(p) \geq 0\}\, k_i(p)\, dF(i),$$

with $K(p) = D(p)(1 - r(p))$.

Welfare for incidence (local). By the envelope theorem,

$$CS'(p) = -K(p) = -D(p)\big(1 - r(p)\big).$$

This replaces the textbook $dCS = -D(p)\, dp$. For local incidence, (D, r) are sufficient; for global DWL they are not.

Implication for incidence under returns. In the perfect-competition notes, the local consumer burden of a price change should use kept volume $K = D(1 - r)$, not gross purchases $Q = D$:

$$\frac{dCS}{dx} = -K(p)\, \rho_x = -D(p)\big(1 - r(p)\big)\, \rho_x, \qquad x \in \{c, h\}.$$

Pass-through formulae (market clearing in μ) are unchanged; only the consumer-side incidence weight is corrected.